Data Collection and Preprocessing Phase

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Project Title: AI-Enhanced Intrusion Detection System
Maximum Marks: 2 Marks

Data Quality Report Template

Data Source	Data Quality Issue	Severity	Resolution Plan
UNSW-NB15 Dataset (CSV File)	Missing or null values in certain features	Moderate	Apply data cleaning techniques such as filling missing values using mean/median or removing incomplete records
Network Traffic Features	Imbalanced dataset (more normal traffic than attack data)	High	Use techniques like SMOTE (Synthetic Minority Oversampling) or resampling to balance the dataset
Feature Encoding	Categorical data not properly encoded	Moderate	Convert categorical features into numerical values using Label Encoding or One-Hot Encoding
Data Normalization	Features with different scales affecting model performance	High	Apply normalization or standardization techniques (Min-Max Scaling or StandardScaler)
Training Data Labels	Incorrect or inconsistent labeling of attack types	High	Validate dataset labels and correct inconsistencies before training
Real-Time Input Data	Incomplete or invalid input provided by user	Moderate	Implement input validation checks in backend before prediction
Model Input Features	Missing required features during prediction	High	Ensure all required input fields are provided and match training dataset structure
Dataset Noise	Presence of irrelevant or noisy data	Low	Apply feature selection and filtering techniques to improve model accuracy
Model Overfitting	Model performs well on training but poorly on new data	High	Use cross-validation, regularization, and proper train-test split
Data File Handling	Incorrect file paths or failure to load dataset	Low	Use proper file handling methods and verify file paths before execution

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Summary

The data preprocessing phase plays a crucial role in ensuring the effectiveness of the AI-Enhanced Intrusion Detection System. By identifying and resolving data quality issues such as missing values, imbalance, and improper encoding, the system can achieve higher accuracy and reliability. Proper preprocessing ensures that the machine learning model performs efficiently in detecting and classifying network intrusions in real-time.
